

MELODY YIYUAN ZHANG

Ph.D. Candidate, University of Michigan, Ann Arbor, MI 48105

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EDUCATION

Ph.D. in Chemical Engineering and Scientific Computing	<i>Expected May 2027</i>
University of Michigan, Ann Arbor, MI	GPA: 3.86
B.S. in Chemical Engineering	<i>May 2022</i>
Purdue University, West Lafayette, IN	GPA: 3.82

AWARDS

Empowering Research with AI Award by the AI Institutes at Michigan (AIIM)	2026
Best Overall Poster Award at U-M Annual Data Science & AI Summit	2025
Michigan Institute for Computational Discovery & Engineering (MICDE) Fellowship	2025
2nd place in Business+Tech Datathon	2025
1st place in College of Engineering, Undergraduate Research Conference Award	2021
Research Grant from Office of Undergraduate Research	2021
Purdue Women in Engineering Program Scholarship	2021
Chemical Engineering Stephen Craig Outstanding Junior Award	2021
AstraZeneca Rising Star Award for initiative, innovation, and productivity	2021
Best Data & Evidence award, Recycling Bin Separation Design	2018

RESEARCH EXPERIENCE

Graduate Student Research Assistant , Lindsey Group & Glotzer Group	<i>Oct. 2022 - present</i>
- Extended the machine-learned interaction model (ML-IAM), ChIMES, to describe how effective interactions between polymer-grafted nanoparticles depend on key physicochemical attributes, and integrated the model with digital alchemy inverse design for efficient optimization of particle attributes towards target self-assembled structures. - Designed a generalized ML-IAM framework that models anisotropic interactions for nanoparticles with convex shapes, enabling broad applicability to nanomaterial systems. - Developed an adaptive parallel meshing algorithm for training data generation that reduces required data volume by $\sim 100\times$ by focusing on key features of the target potential energy surface. - Contributed to the development and improvement of state-of-the-art machine-learning and simulation software, including ChIMES, HOOMD-Blue, and Freud.	
Undergraduate Research Assistant , Dou Group, Purdue University	<i>Jan. 2020 - May 2022</i>
- Modified the surface of formamidinium-based perovskites using synthesized ligands to passivate defects and kinetically trap the photovoltaic crystal phase. - Incorporated amine-thiol chemistry in the perovskite layer and achieved a power conversion efficiency (PCE) of 20.1% and improved stability under ambient conditions. - Streamlined the data analysis process of photovoltaic device characteristics and performance.	

RESEARCH PUBLICATIONS

Zhang, M. Y., Lee, S. K. A., Glotzer, S., and Lindsey, R. A Generalized Machine-Learning Interaction Model for Anisotropic Interactions in coarse-grained nanoparticles. Manuscript in preparation.

Zhang, M. Y.*, Lee, S. K. A.*, Glotzer, S., and Lindsey, R. A Generalized Machine-Learning Framework for Developing Alchemical Many-Body Interaction Models for Polymer Grafted Nanoparticles. *J. Chem. Theory. Comput.* 2025. DOI: 10.1021/acs.jctc.5c00901 *equal contribution

Finkenauer, B., **Zhang, Y.**, Ma, K., Turnley, J.W., Schulz, J. et al. Amine-Thiol/Selenol Chemistry for Efficient and Stable Perovskite Solar Cells. *The Journal of Physical Chemistry C*. 2023, 127, 2, 930-938. DOI: 10.1021/acs.jpcc.2c06967

Zhang, Y. Interface Engineering to Kinetically Trap the Photovoltaic Crystal Phase of Formamidinium Lead Iodide Perovskite. *The Journal of Purdue Undergraduate Research*. Vol. 11, Article 15. (2021). DOI: 10.7771/2158-4052.1533

CONFERENCE PRESENTATIONS

Zhang, M. Y., Lee, S., Glotzer, S. C., Lindsey, K. R. A Generalized Framework for Developing Alchemical Coarse-grained Interaction Models for Self-assembling Polymer-grafted Nanoparticles. Oral Presentation. 2025. *Midwest Thermodynamics and Statistical Mechanics Conference*.

Zhang, M. Y., Lee, S., Laubach, B. Glotzer, S. C., Lindsey, R. Generating Nanomaterial Design Surfaces with Physics-Informed Machine Learning Models. Oral Presentation. 2024. *American Institute of Chemical Engineers*.

Zhang, M. Y., Lee, A., Glotzer, S. C., Lindsey, R.K. Towards Mapping Nanomaterial Design Space with Physics-Informed Machine-Learned Models. Poster Presentation. 2024. *Midwest Thermodynamics and Statistical Mechanics Conference*.

Zhang, M., Lee, A., Laubach, B., Tsai, S. Anderson, J.A., Glotzer, S. C., Lindsey, R.K. A machine-learning strategy for the physics-informed design of nanomaterials using ChIMES. Poster Presentation. 2024. *American Physics Society*.

PROFESSIONAL EXPERIENCE

Graduate Student Instructor, Mathematical Analysis for Chemical Engineers *Aug. - Dec. 2025*

- Developed and presented detailed homework solutions and lectures, consistently receiving strong student and instructor evaluations.
- Provided critical support through office hours and oversaw exam administration and logistics.

Validation Engineering Intern, AstraZeneca, Mt. Vernon, IN *May - July 2021*

- Employed the BIOVIA Discoverant data analysis software to compile an annual product quality report on two extensively produced active pharmaceutical ingredients in the company.
- Designed a computer application using Tulip to provide intuitive instructions for equipment and process risk assessments for the engineering and operation teams.

Resident Assistant, Purdue University *Nov. 2020 - May 2022*

LEADERSHIP EXPERIENCE

Liaison to the Director, Center for Simulation and Data Intensive Research in Chemical Engineering (CSiDIR-CHE), University of Michigan *2023 - 2024*

Engineering Peer Mentor, Women in Engineering Program, Purdue University *2020*

Engineering Peer Mentor, Boiler Student Network, Purdue University *2020*

Events Planning Chair, UR Multicultural Connections, Purdue University *2020 - 2021*

Purdue AIChE Officer, Student Relation & Faculty Relation Committee *2020*

Vice President, Wiley Resident Hall Club, Purdue University *Aug. 2019 - 2020*

RELEVANT COURSEWORK

Design and Analysis of Algorithms	Simulation of Condensed Matter
Machine Learning	Statistical & Irreversible Thermo
Parallel Computing	Applied Math for Chemical Engineering
Numerical Linear Algebra	Big Data Management Tools

PROFESSIONAL SKILLS

Programming Languages and Software: Python, C++, Bash, Git, Blender, MPI, OpenMP, CUDA

Machine Learning: Variational Autoencoder, SBERT, Graph Neural Networks, Scikit-learn, Pytorch

Molecular Simulation: LAMMPS, HOOMD-blue, OVITO, VMD